Worksheet 6.5

The big bang

Student: .... Class: ....

Flow charting the big bang

1. Complete the big bang flow chart by selecting the event from the list below that matches each step in the sequence and rewriting it in the correct box. Enter a title for each box that summarises each event.

- $t = 0$
- $t = 10^{-43}$ s
- $t = 10^{-34}$ s
- $t = 10^{-4}$ s
- $t = 10^{-2}$ s
- $t = 1$ s
- $t = 5$ min
- $t = 3 \times 10^{5}$ yr
- $t = 2 \times 10^{8}$ yr
- $t = 1 \times 10^{9}$ yr

Big bang events in random order:

A. Universe has cooled to 3000 °C; first atoms form.
B. Galaxies begin to form.
C. Time and space form; space expands rapidly.
D. Heavier matter forms — protons, neutrons.
E. Singularity exists — the big bang occurs.
F. Universe cools; reaches size of the solar system.
G. Atomic nuclei form.
H. First stars appear.
I. Light matter forms — electrons, positrons.
J. Universe has cooled to 10^{10} °C.

Evidence for the big bang

2. The following microwave map of the early universe (t = 380 000 years after big bang) was produced by a probe that orbited the Earth in 2001.

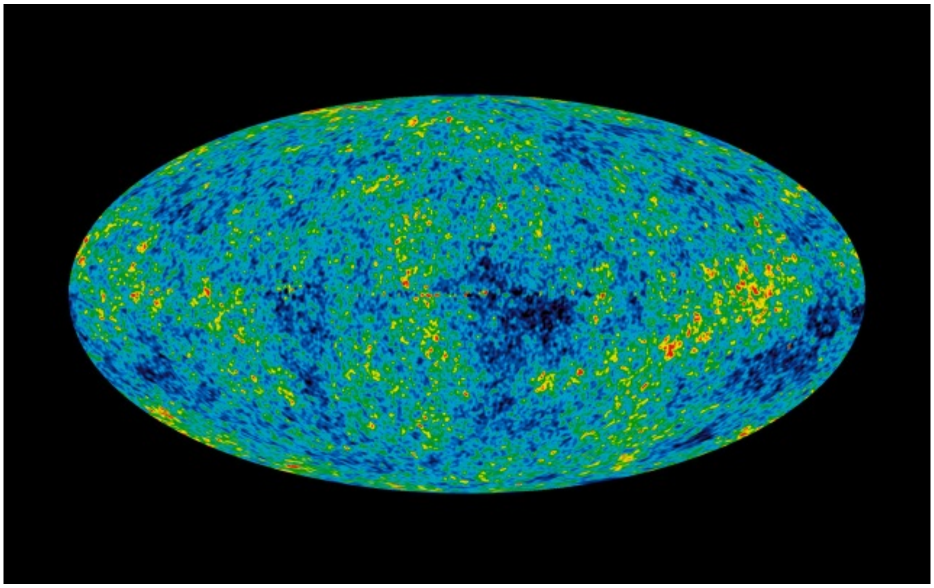

Photo: NASA

(a) The map shows that in the early universe matter was not uniformly distributed. Explain how this map supports the big bang theory.

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(b) In 1948, Gamow and Alpher predicted that microwave radiation should now fill the universe. They proposed that this radiation was the cooled 'afterglow' of the big bang. When was this microwave radiation first detected?

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